

ISAAC (ZAC) ADJEI

✉ contact@isaacadjei.me | 🌐 www.www.isaacadjei.me | [LinkedIn](#) | [GitHub](#)

PROFILE

Software Engineering student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026. Builds production-grade full-stack web applications with modern frameworks, cloud infrastructure and secure authentication. Seeking software engineering internships and industrial placements.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C, C++, Java
Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Framer Motion
Backend: Node.js, FastAPI, PostgreSQL, Redis, REST APIs, JWT Auth, RBAC
Cloud Devops: AWS, Docker, Vercel, Cloudflare, GitHub Actions, CI/CD, Terraform
Tools: Git, GitHub, VS Code, Postman, Figma

PROJECTS

PHAEMOS: Smart Maintenance Platform 2026, ongoing

Next.js 14, FastAPI, PostgreSQL, Redis, Docker, ESP32, Arduino, STM32, scikit-learn

- End-to-end IoT predictive maintenance platform with ESP32, Arduino and STM32 nodes
- FastAPI backend processing 1000+ telemetry readings/day with sub-200ms response times
- Isolation Forest ML anomaly detection triggering automated maintenance tickets

[Website ↗](#) | [GitHub ↗](#)

Personal Portfolio & Dashboard 2025-2026

Next.js 14, React, TypeScript, Tailwind CSS, Supabase, PostgreSQL

- Server-side rendered portfolio with dynamic OG image generation
- Private dashboard with goals tracking, job applications, habits and encrypted vault
- Implemented JWT authentication, audit logging and role-based access control

[Website ↗](#)

Embedded Sensor Networks 2025

STM32, Arduino, C++, I2C, SPI, MQTT, InfluxDB

- Multi-node sensor network collecting environmental data (temp, humidity, pressure)
- STM32 master coordinating 8+ slave nodes via I2C with error handling
- Real-time data streaming via MQTT to InfluxDB time-series database

EDUCATION

Aston University, Birmingham Oct 2024 - Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist - Black Heritage Undergraduate of the Year Award 2026

Relevant Modules: Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Digital Circuit Design, Object-Oriented Programming (C++)

EXPERIENCE

TargetJobs / Sky 2026

Top 40 Finalist - Black Heritage Undergraduate of the Year

- Recognised among thousands of applicants for academic excellence and leadership potential
- Assessment centre experience including case studies and technical interviews

Cancer Research UK 2022-2024

Shop Volunteer

- Customer service and retail operations in charity shop environment
- Cash handling, stock management and visual merchandising